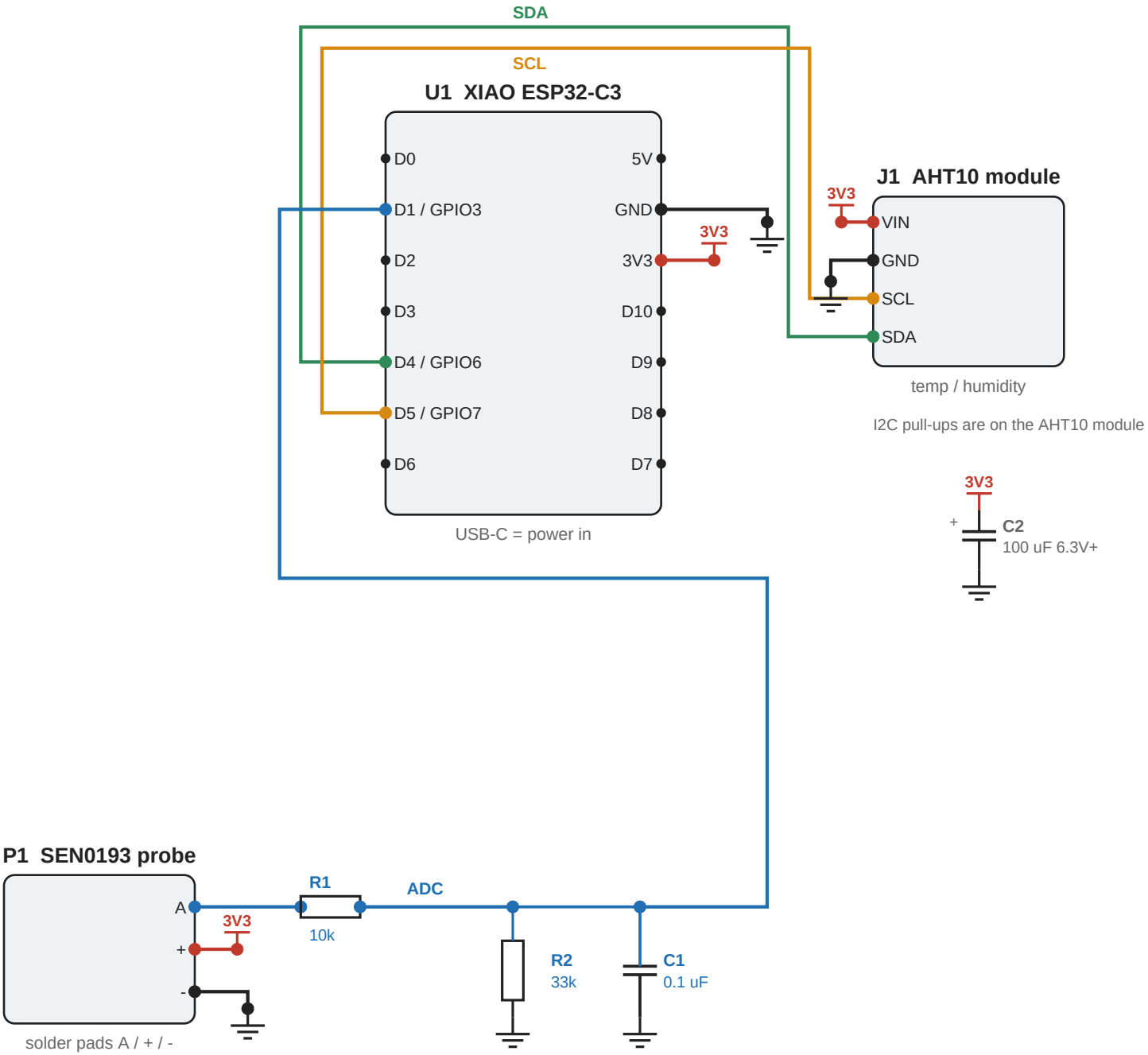
ESP-Soil Stake v1 — Schematic

One stake per pot: XIAO ESP32-C3 + AHT10 + SEN0193 probe. USB-C power, Wi-Fi to Prometheus/Grafana.



Why the divider (R1/R2)

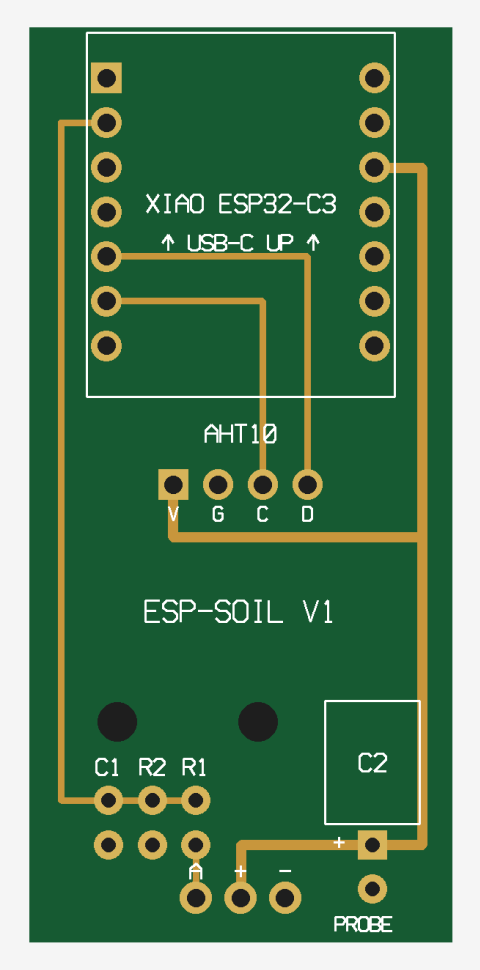
The ESP32-C3's ADC tops out around 2.5 V. A dry SEN0193 can sit right at that, so R1/R2 scale it by 0.767 (2.5 V -> 1.92 V, 1.2 V -> 0.92 V). Calibration uses the voltage at the ESP pin, so nothing else changes. Don't want it? Fit a wire link in R1 and leave R2 empty.

Nets

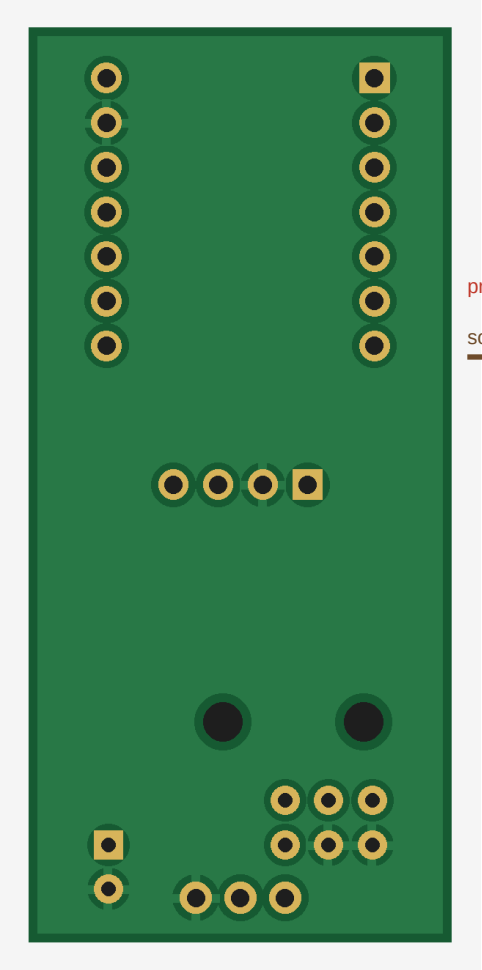
ADC: P1.A -> R1 -> D1/GPIO3 (R2 + C1 to GND) SDA: D4/GPIO6 SCL: D5/GPIO7 3V3: XIAO 3V3 -> AHT VIN, P1 +, C2

Board Layout and Stake Assembly

24 x 52 mm, 2-layer, 1.6 mm. Top: all signals. Bottom: solid ground pour.

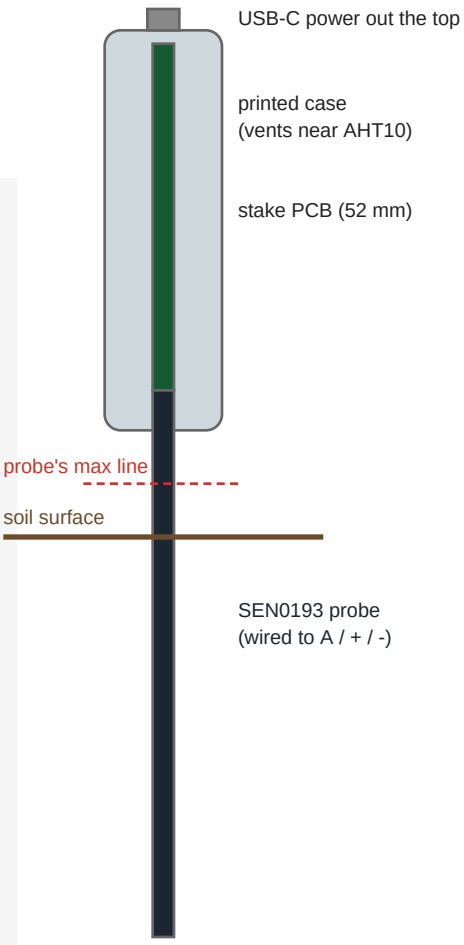


TOP



BOTTOM (mirrored view)

Stake side view



Seal it

Case must seal at the probe (silicone or a gland). Only the probe below its line goes in soil. Conformal-coat the probe's electronics and this board.

Mechanical

Board 24 x 52 mm, same width as the SEN0193. USB-C sits at the top edge (y = 0). Two M2 holes 8 mm apart at 39.5 mm down for screwing the board to the case. Tallest parts: XIAO on headers (~8 mm) and the AHT10 module. Lay C2 (electrolytic) flat over its outline. Keep the AHT10 near case vents and away from the XIAO for honest readings.

Order it

Upload esp-soil-stake-gerbers.zip to JLCPCB or PCBWay: 2 layers, 1.6 mm, HASL. Defaults are fine; check the preview in their viewer before paying. 5 boards is usually a few dollars plus shipping.

BOM, Assembly and Calibration

All through-hole. Hand-solderable.

Ref	Part	Qty/board	Notes
U1	Seeed XIAO ESP32-C3	1	Solder on 1x7 male headers (or flush)
J1	AHT10 module (VIN GND SCL SDA)	1	Check your module's pin order matches V G C D
P1	DFRobot SEN0193 probe	1	Remove its JST plug, wire 3 short leads to A + -
R1	10k resistor, 1/4W	1	Or a wire link to skip the divider
R2	33k resistor, 1/4W	1	Leave empty if R1 is a link
C1	0.1 uF ceramic, 2.54 mm	1	ADC noise filter
C2	100 uF 6.3V+ electrolytic, 2.5 mm	1	Lay flat, + to square pad
—	2x 1x7 female headers (optional)	1	Lets you unplug the XIAO; adds height
—	2x M2 screws	2	Mount to case

Assembly order

1. Solder R1, R2, C1 first (lowest parts), then C2 bent flat.
2. Power test the bare board: plug nothing in, check for shorts with a meter (3V3-GND, 5V-GND should NOT beep).
3. Solder the AHT10 header and module.
4. Solder the XIAO last. Square pad = D0 (top left), USB-C toward the top edge.
5. Wire the probe: A = signal (blue), + = 3V3 (red), - = GND (black). Keep leads short.
6. Flash soil-stake.yaml over USB (change plant / node_name per stake), then mount in the case.

Calibration (per stake)

Watch 'Plant N raw' in Grafana/HA/logs. This is the voltage at the ESP pin, after the divider.
dry_v = probe in air or bone-dry mix. wet_v = 30 min after watering to runoff. Put both in the YAML and reflash.

Grafana

Add each stake's IP:80 to prometheus.yml. Sensors are named 'Plant N moisture / temperature / humidity / leaf VPD', so the existing dashboard picks them up. Give each stake a fixed IP in UniFi.

Heads-up

The AHT10 now sits at soil level, not canopy height, so expect warmer/more humid readings than at the canopy.
The ESP warms the board slightly; compare against a trusted thermometer once and set the offset in the YAML.
Unverified: this layout was generated without KiCad, so check the Gerbers in JLCPCB's viewer and measure your XIAO and AHT10 against the footprint (XIAO rows 15.24 mm apart, 2.54 mm pitch) before ordering.